

Finding cells that do not belong

How AskCell detects an abnormal cell population by comparing a specimen against healthy people — and the one thing that makes it hard.

AskCell v1.0 flow cytometry synthetic-data results

1 The question

A flow cytometer takes a bone-marrow sample, tags the cells with glowing antibodies, and pushes them past a laser one at a time. For every cell it records how much of about fourteen surface proteins that cell carries. The output is a file that is, in plain terms, a spreadsheet: one row per cell, one column per protein, a few hundred thousand rows.

The question AskCell answers is:

Does this specimen contain a population of cells that does not exist in healthy marrow — and if so, how large is it, and what does it resemble?

Notice what that is not. It is not "does this patient have cancer." A diagnosis in blood cancer needs cell shape under a microscope, chromosome analysis, and molecular tests as well. Two specimens can look identical to a cytometer and be different diseases. Keeping those two questions apart is the difference between a useful tool and a misleading one.

2 Why it does not learn from cancer

The obvious approach would be to show a model thousands of cancer samples and thousands of healthy ones and let it learn the difference. AskCell deliberately does not do this.

A model trained that way learns the particular diseases in its training set. Show it a leukaemia subtype it has never seen and it has no reason to react. It has learned a list, not a principle.

Instead AskCell learns only what *healthy* looks like, and reports anything that departs from it. That generalises to abnormalities nobody has catalogued, because it never needed to know what it was looking for. The cost is that it cannot name what it found — which is handled separately, in section 7, and with much weaker claims.

There is a practical benefit too. Labelled cancer data is scarce and hard to get. Healthy samples are not. An approach that needs only healthy reference material is one that a project without a hospital behind it can actually attempt.

3 Defining normal

Every cell becomes a point in thirteen dimensions — one per protein measured. Pool the cells from several healthy people and you have a description of every cell type healthy marrow contains, in roughly the right proportions.

To ask whether a patient's cell is normal, find its nearest neighbours among the healthy cells and measure the average distance. A cell sitting inside a healthy population has close neighbours and scores low. A cell whose combination of proteins exists in nobody healthy has none, and scores high.

Where the cutoff comes from

Those scores are distances, so "far" means nothing without a scale. Picking a cutoff by hand would be guessing, and a guessed threshold is the easiest way to build something that looks excellent on its own test data.

So the cutoff is *measured*. Each healthy person is scored against a reference built from only the *others* — which shows how unusual a healthy stranger looks. The cutoff is set at the 99.9th percentile of that spread.

This makes the false-alarm rate a chosen quantity rather than a surprise: about one healthy cell in a thousand will trip the flag, by design.

It also means a single healthy sample cannot build a reference. With nothing to hold out there is nothing to measure the cutoff against, so the tool refuses rather than guessing. Two is the arithmetic minimum; four to six is comfortable.

4 Why one stage is not a detector

Follow the previous section to its conclusion. If the cutoff admits one healthy cell in a thousand, then on a file of 400,000 cells it will flag several hundred cells *in a completely healthy person*.

That rate cannot be tuned away. Raising the cutoff until healthy samples come back clean also raises it past the small populations that matter most — the faint traces of disease returning after treatment, which are the whole reason for testing repeatedly.

So flagging distant cells is only the first stage. The real question is not "are there odd cells" but "**are the odd cells many, and all odd in the same way?**"

That question has a geometric answer. Cancer begins as one cell that stops maturing and starts copying itself, so its descendants are near-identical and land in a tight knot. Spurious flags come from unrelated cells in sparse corners of the data, so they scatter with nothing near them. Identical scores, completely different shapes.

cells in the specimen	372,650
stage 1 – flagged as distant	531 (0.142%)
discarded as scattered	342
stage 2 – clustered together	189 (0.051%)
true answer	191 (0.051%)

Stage 1 alone would have overstated the disease by nearly threefold. On a healthy specimen, 54 cells were flagged, all 54 were scattered, and nothing survived stage 2 — correctly reported clear.

Both numbers appear in the report. The gap between them is the noise the clustering removed, and showing it is what lets someone check the result instead of trusting it.

5 The thing that makes this hard

Healthy bone marrow is a factory. It is constantly manufacturing new immune cells, and its young B-cell precursors — called *hematogones* — carry almost exactly the surface proteins of a leukaemia cell. They are abundant in healthy children and they multiply after chemotherapy.

A detector that asks "are there cells with CD19, CD10, and dim CD45?" reports leukaemia in healthy people. This is not a hypothetical risk; it is what the obvious first implementation does.

What separates them is not any protein. It is structure.

Young cells are on a journey. Catch a crowd of them and you see every stage at once, spread smoothly along the path — and the proteins move together, one fading as another brightens, because the cells are maturing in order.

Cancer cells are stuck. All frozen at the same moment, all near-identical, because they are copies of one cell that stopped growing up.

Normal is a smooth spread along a path. Cancer is a tight clump.

This is why the synthetic test data deliberately includes hematogones, generated along a correlated maturation path, and why two of the tests *require* that no single protein can separate them from cancer cells. Building against easier fake data would produce something that scores beautifully in testing and collapses on the first real patient.

It also means specificity — not raising a false alarm — has to be reported alongside sensitivity, and the healthy controls have to contain the look-alikes, or the specificity figure is worthless.

6 What the measurements showed

Nineteen benchmark runs on synthetic specimens with known answers, at three different acquisition depths.

sensitivity, detected cases	97.9%
precision, detected cases	100%
healthy controls cleared	4 / 4
repeat runs	identical
speed	1.5 s per 100k cells

The last figure is worth pausing on. A trained specialist doing this by hand takes thirty to sixty minutes per specimen, drawing shapes with a mouse on two proteins at a time. There are fourteen, and no human can see fourteen dimensions at once. That is not a claim about being cleverer than a specialist; it is a limitation of two-dimensional plots that a computer simply does not have.

The result I did not expect

50,000 cells acquired	found 0.1% ≈ 50 cells
200,000 cells acquired	found 0.05% ≈ 100 cells
500,000 cells acquired	found 0.01% ≈ 50 cells

Across a tenfold range of depth, the smallest findable population stayed at roughly fifty to a hundred cells — while the *percentage* it represents moved by a factor of ten.

The limit is a cell count, not a percentage. The detector needs a certain number of cells before a group is recognisable as a group. What fraction of the specimen that represents is not a property of the software at all — it is set by how many cells went through the machine.

So the way to detect a smaller percentage is to acquire more cells. This is the same tradeoff real residual-disease tests make, and the reason they run millions of cells rather than thousands. Quoting "0.01%" without saying at what depth is the most flattering and least informative way to state it.

7 Naming what was found

Once a population is found, its protein pattern can be compared against the published patterns of known conditions. The report ranks the candidates, with the evidence for and against each, and the tests that would actually settle it.

This part makes deliberately weak claims. Surface proteins alone cannot classify a blood cancer — so the wording is always that a pattern *resembles* something, never that it is something, and every candidate arrives with the confirmatory tests it would need.

Ranking on proteins alone had an instructive failure. On a textbook cancer clone it put *hematogones* — the benign look-alikes from section 5 — in first place. That was not a tuning error. The two share a protein pattern closely enough that no weighting of proteins separates them, so a protein-only ranking will confidently call a leukaemia normal.

The fix was to give each candidate an expected *shape*: a clone should be tight, maturing cells should be spread out. With that in place, the same protein values produce opposite conclusions depending on the geometry — which is correct. The benign candidate is pushed down but kept visible, because "this is what it could be mistaken for" is worth knowing.

8 What would change the answer

The honest weaknesses, in order of how much they matter.

- **Every number here is on synthetic data.** The generator encodes one idea of what an abnormal population looks like, so these figures measure internal consistency, not clinical accuracy. Real specimens are messier in ways a generator does not know to imitate. No real specimen has been tested.
- **One parameter was hand-tuned** — how close two cells must be to count as clumped together. It was set by measuring the gap between real clumps and scattered noise on the test fixtures, which is tuning against test data. It is expressed as a multiple of a quantity measured from the reference rather than a fixed number, and a guard test re-checks that the gap still exists, but it needs re-measuring on real data.
- **The reference must match the specimen exactly.** Protein intensities are not comparable across different antibody panels, instruments, or even detector settings. A reference built one way cannot judge a specimen acquired another way, and the tool refuses rather than approximating.
- **A small or unrepresentative reference is dangerous.** A rare-but-normal cell type missing from the reference will be flagged in every single patient.
- **Anything under about thirty cells is refused** as indistinguishable from a chance clump. The remedy is acquiring more cells, not loosening the rule.

9 What is still needed

Two questions block everything else. The rest can proceed on assumptions and be corrected later.

- **Which proteins does the machine actually measure, and with which dyes?** This defines the panel, and the panel defines the entire data model. The current test fixtures assume a fourteen-colour leukaemia panel, which is an educated guess.

- **How many healthy samples can be acquired on that panel?** Without healthy reference material on the matching panel there is nothing to compare against, and the tool cannot run on real data at all.

Beyond those: whether any specimens exist with a confirmed diagnosis (the only path to an accuracy claim), how many cells a routine run acquires (which sets the honest detection limit), and whether instrument settings are saved with each file (without which drift between sessions cannot be separated from biology).

– Words worth knowing

Event

One cell measured by the machine. One row of the table.

Marker

A protein on the cell's surface, detected with a dye-tagged antibody. CD19, CD10 and CD45 are markers.

Panel

The specific set of markers a lab chose to measure, plus the dyes and instrument used. Not interchangeable between labs.

Gating

Selecting a subset of cells for analysis. Traditionally done by hand, drawing shapes on a two-dimensional plot.

Blast

An immature cell. The percentage of blasts is how acute leukaemia is defined and graded.

Hematogone

A normal young B-cell in marrow whose markers closely resemble a leukaemia cell. The central difficulty — section 5.

Clonal

Descended from one cell, therefore near-identical. Shows up as a tight cluster.

MRD

Measurable residual disease — the small amount of cancer left after treatment. What repeat testing hunts for, and where detecting tiny populations matters most.

Sensitivity

Of the cells that truly were abnormal, the fraction found.

Precision

Of the cells called abnormal, the fraction that truly were.

Specificity

Of the healthy specimens, the fraction correctly reported clear.

Immunophenotype

The full pattern of surface markers on a group of cells.

Data leakage

Information reaching a model that it would not have in real use, producing performance that is not real.

Research and educational use only — not a diagnostic device, and not clinically validated. Every figure above is measured on synthetic data and describes internal consistency rather than clinical accuracy. Condensed from the full technical proposal, which carries the module architecture, acceptance criteria and validation protocol in full.